EXP 3. Water Jug Problem

Aim

To measure exactly 2 litres using a 4-litre jug and a 3-litre jug.

Algorithm

1. Represent each state as (amount in 4-litre jug, amount in 3-litre jug).
2. Start with state (0, 0) and place it in a queue.
3. Generate states by filling a jug, emptying a jug, or pouring water between jugs.
4. Use a parent dictionary to avoid revisiting states and record the path.
5. Stop when the 4-litre jug contains exactly 2 litres.
6. Print the sequence of states from the initial state to the goal.

Program

*AI.py - C:/Users/Barath/AppData/Local/Programs/Python/Pytho

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```
from collections import deque

start = (0, 0)
goal = 2
q = deque([start])
parent = {start: None}

while q:
    a, b = q.popleft()
    if a == goal:
        path = []
        s = (a, b)
        while s is not None:
            path.append(s)
            s = parent[s]
        for state in reversed(path):
            print(state)
        print("Goal Reached")
        break

    pour_ab = min(a, 3 - b)
    pour_ba = min(b, 4 - a)
    states = [
        (4, b), (a, 3), (0, b), (a, 0),
        (a - pour_ab, b + pour_ab),
        (a + pour_ba, b - pour_ba)
    ]

    for state in states:
        if state not in parent:
            parent[state] = (a, b)
            q.append(state)
```

Output

```
==== RESTART: C:/Users/Barath/AppData/Local/Programs/Python/Python313/AI.py ====
(0, 0)
(4, 0)
(1, 3)
(1, 0)
(0, 1)
(4, 1)
(2, 3)
Goal Reached
```

Result

Thus, exactly 2 litres were measured in the 4-litre jug using the water jug operations.